## Assignment: Diabetes Prediction using Machine Learning

In this assignment, you will build a **Machine Learning classification model** using the provided **Diabetes Dataset** to predict whether a patient is likely to have diabetes based on various medical attributes.

Your goal is to implement an end-to-end **Machine Learning classification pipeline**, including data exploration, preprocessing, feature engineering, model building, training, evaluation, and performance analysis using **Python and Scikit-learn**.

This assignment will help you understand:

- End-to-end **Machine Learning workflow**
- Exploratory Data Analysis (EDA)
- Data preprocessing and handling missing values
- Feature selection and feature engineering
- Splitting data into training and testing sets
- Building and training **classification models**
- Understanding classification algorithms such as **Logistic Regression**
- Model evaluation using metrics such as **Accuracy**
- Interpreting Machine Learning model results and drawing business/analytical insights

**Dataset Description**

The dataset contains patient health-related attributes that can be used to predict the likelihood of diabetes.

Some important columns include:

| Column | Description |
|---|---|
| Pregnancies | Number of times pregnant |
| Glucose | Plasma glucose concentration |
| BloodPressure | Diastolic blood pressure |
| SkinThickness | Triceps skin fold thickness |
| Insulin | Serum insulin level |
| BMI | Body Mass Index |
| DiabetesPedigreeFunction | Diabetes pedigree score |
| Age | Age of the patient |
| Outcome | Target Variable (0 = No Diabetes, 1 = Diabetes) |

## Assignment Tasks

You must complete the following stages as part of this Machine Learning assignment.

## Part 1 — Data Preparation

Prepare and preprocess the dataset for Machine Learning model training.

### Tasks

1. Load the dataset using **Pandas**.
2. Explore the dataset:
    a. Shape of the dataset
    b. Column names
    c. Data types
    d. Missing values, if any
    e. Basic statistical summary
3. Separate the dataset into:
    **a. Features (X)**
    **b. Target Variable (y)**
4. Perform a **Train-Test Split**.
5. Perform appropriate preprocessing, if required:
    a. Handle missing values
    b. Encode categorical variables, if any
    c. Scale/standardize numerical features where appropriate

### Deliverables

- Dataset exploration summary
- Description of features and target variable
- Preprocessing steps performed
- Training and testing dataset sizes

## Part 2 — Machine Learning Model Building

Build a suitable **classification model** using Scikit-learn.

### Tasks

1. Import the required **Python and Scikit-learn libraries**.
2. Select and implement an appropriate classification algorithm, such as Logistic Regression
3. Train the selected Machine Learning model using the training dataset.
4. Explain why the selected algorithm is suitable for this problem

### Deliverables

- Machine Learning model(s) implemented
- Explanation of the selected algorithm

### Part 3 — Model Training

Train the Machine Learning model using the prepared training dataset.

### Tasks

1. Fit the selected model using the training dataset.
2. Generate predictions on the training dataset.
3. Evaluate the model's performance – accuracy

### Deliverables

- Model training output
- Training accuracy
    a.
2. Compare the training and testing performance of the model.

### Deliverables

- Test accuracy
- Precision, Recall, and F1-Score
- Confusion Matrix
- Comparison of training vs. testing performance
- Brief interpretation of the model results

### Part 4 — Model Performance Analysis

Analyze the overall performance of the Machine Learning model and draw meaningful conclusions.